Kalshi Market Making Algorithm

George Lord, Chris Mulligan, and Max Zhalilo
(12243747, 12502987, 12340701)

February 28, 2026

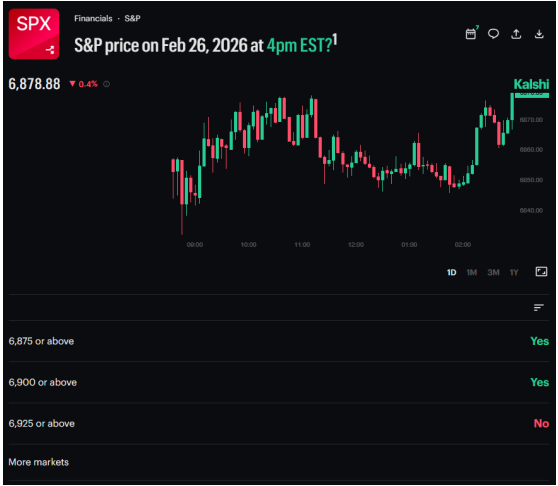
Agenda

1. Overview
2. Kalshi SPX Event Contracts
3. Replication by Binary Calls/Puts
4. Modeling SPX Price CDF Log-Normally
5. Kalshi Market Making from Fair Prices (VIX-based Spread)
6. Delta-Hedging Kalshi Contracts with SPY
7. Steps Together
8. Required Data and Computations

Overview

- We run automated market-making strategies on short-term (0-1 Day) SPX Price Events.
- We model the probability of SPX being above/below a price at a given time log-normally, using live SPX and VIX.
- We use this to fair price binary calls/puts, then fair price Kalshi SPX events via replication.
- We also use VIX to adjust position sizing.
- We hedge our delta exposure with SPY.

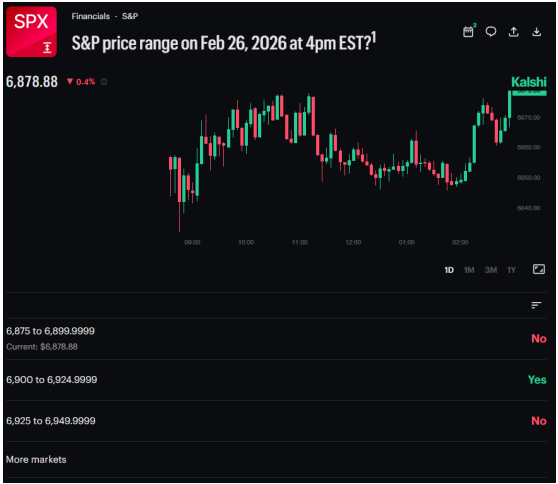
Kalshi Contract 1: KXINXU



The KXINXU market for Feb 26, 2026, which has resolved

- KXINXU (as prefixed by Kalshi API) contracts bet whether the end-of-day S&P 500 index value will be above a given value after a specific day and time in the future (as measured by SPX value)
- "Yes" and "No" contracts at various index values are actively traded 24/7 leading up to the last trading day, after which contracts are paid out. Each contract for a correct prediction pays out $1, and wrong predictions pay out $0
- As such, "Yes" contracts act as binary calls on SPX, and "No" contracts as binary puts, for a given strike price (the "above" price, e.g. $6,875) and expiration date (2/26/26 4PM EST in this example)

Kalshi Contract 2: KXINX



The KXINX market for Feb 26, 2026, which has resolved

- KXINX contracts bet whether the end-of-day S&P 500 index value will fall in a specific range after a specific day and time in the future (as measured by SPX value)
- "Yes" contracts can be replicated as a position long one binary call with strike at the bottom of the range, and short one binary call with strike at the bottom of the range (binary call spread). "No" contracts would be replicated by going long one binary put at the bottom of the range, and long one binary call at the top of the range (binary strangle).

Binary Call/Put Risk-Neutral Pricing

Recall that a binary call on S expiring at time T with strike K has risk-neutral time $t < T$ price of:

$$V_t = e^{-\int_t^T r(u)du} \mathbb{Q}_t(S_T > K)$$

where

- $e^{-\int_t^T r(u)du}$ is the discount factor based on risk-free rate r
- $\mathbb{Q}_t(S_T > K)$ is the risk-neutral probability of S finishing over K at time T (based on all info at time t)

Similarly, a binary call has a risk-neutral price equal to the discounted risk-neutral probability of S finishing under K at time T.

Replication

Key Observation

All of these Kalshi contracts can be replicated with (combinations of) binary calls/puts

Implications

- At 0 to 1 DTE, binary call/put prices approach the probability of SPX ending above/below a given price (discount $\to 1$ as $T - t \to 0$)
- If we know/can accurately estimate these probabilities, we can fairly price binary calls/puts on SPX, and therefore (via replication) fairly price KXINXU and KXINX for a given SPX price/range and expiration

Modeling SPX Price CDF Log-Normally

Remaining consistent with the typical Black-Scholes (lognormal) model for binary calls/puts, we can model the required probabilities as:

$$\mathbb{P}_t(S_T > K) = \Phi(d_2(K)) \text{ and } \mathbb{P}_t(S_T < K) = 1 - \mathbb{P}_t(S_T > K)$$

$$d_2(K) = \frac{\ln(F/K) - \frac{1}{2}\sigma^2(T-t)}{\sigma\sqrt{T-t}}$$

where

- $F \approx S_t =$ current SPX value (since discount is negligible with 0 to 1 DTE, don't trade over dividends)
- K is the corresponding binary strike price
- T is the corresponding contract resolution date+time, and t the current time, such that $T - t$ is in years (float64)
- σ is assumed flat and taken from current VIX in percent /100

Scaling Market-Making Bid-Ask Spread with VIX

In market making, our position size is directly related to the spread width at which we bid/ask. We have fair values for Kalshi contracts for a given price (range) and time to expiration. We can use this as the center for each contract, and then adjust our width based on VIX. We will test various methods for this as part of our project, but they will generally:

- Use our fair-prices for the contracts as described before as our center
- Calculate our bid-ask spread width as a function of
  - Real-time VIX level (as a measure of volatility)
  - Time to Kalshi Expiration (e.g. time to 2/26/26 4PM EST in prev. examples)
  - Our current net position in YES shares (positive = long YES)
  - Tick size (usually 1 cent)
  - Kalshi's fees (so we don't post bid/asks with -EV after fees)
- We will generally widen as settlement approaches and volatility rises.

Delta-Hedging with SPY

In order to remove directional risk from SPX moves as we accumulate Kalshi positions by market-making (and being filled), we will delta-hedge with SPY. Because our Kalshi contract fair values are a function of current SPX, and since the SPY ETF has (by construction) a delta $\frac{\partial \text{SPY}}{\partial \delta \text{SPX}} \approx \frac{\text{SPY}}{\delta \text{SPX}} \approx 0.1$, we can delta hedge a Kalshi contract by calculating the net delta of it's replication's individual binary call/puts via:

$$\Delta_{\text{YES ABOVE}}(K) = \frac{\partial(\Phi(d_2(S)))}{\partial S} = \frac{\varphi(d_2(S_t))}{S_t \sigma \sqrt{T-t}} \text{ and } \Delta_{\text{NO ABOVE}}(K) = -\Delta_{\text{YES ABOVE}}(K)$$

then summing all our open Kalshi position deltas Δ_i by their quantity q_i as:

$$\Delta_{\text{book,SPX}} = \sum_i q_i \Delta_i$$

which is in dollars of contact value per 1 SPX point. Since 10 SPX point $\approx$ 1 SPY dollars, we take position

$$-10\Delta_{\text{book,SPX}} \text{ shares of SPY}$$

Putting it Together

SPY and VIX Values

↓ Log-Normal CDF

$P_t(S_T > K)$

↓ Risk-Neutral Pricing

SPX Binary Put/Call Fair Price

↓ Replication via Binaries

Kalshi SPX Contract Fair Prices

↓ Spread based on VIX

Market Making on KXINX(U)

↓ Partial of Kalshi Fair Value w.r.t SPX

Delta-Hedging on SPY

Required Data and Computations

(Historical) Data Sources

- 1s OHLCV SPY from Databento
- 1s SPX and VIX values from Bloomberg
- KXINX(U) Contract specs (settlement, "strike" prices, ranges) from Kalshi API
- KXINX(U) Order book updates from Kalshi API
- (Possibly) risk-free rate data for discounting (e.g. SOFR OIS)

Computations

- SPX CDF over historical time-series
- Implied KXINXU binary call/put fair values
- Implied KXINX binary combo fair values
- Our KXINX(U) bid/asks based on fair values, VIX, and Kalshi historical data
- Simulated fills based on Kalshi historical and assumed p_{fill}
- Deltas and delta-hedges based on simulated fills, SPY historical, and fair value partials
- Trades, PnL, and performance metrics based on simulated fills and SPY hedges